CBSE Practice Worksheet

Class 8 | Science | Difficulty: Hard | Set E

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the wavelength of Lyman series limit?
   (A) 91.2 nm
   (B) 121.6 nm
   (C) 656.3 nm
   (D) 364.6 nm
2. What is the oxidation state of Mn in $KMnO_4$?
   (A) +2
   (B) +4
   (C) +6
   (D) +7
3. Which principle explains the working of a hydraulic lift?
   (A) Archimedes
   (B) Pascal
   (C) Bernoulli
   (D) Newton
4. What is the wavelength of de Broglie wave for an electron?
   (A) $\lambda = h/mv$
   (B) $\lambda = mv/h$
   (C) $\lambda = hv/m$
   (D) $\lambda = m/hv$
5. What is the coordination number in FCC structure?
   (A) 6
   (B) 8
   (C) 12
   (D) 14
6. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
   (A) 25%
   (B) 33%
   (C) 50%
   (D) 75%
7. Which enzyme is responsible for DNA replication?
   (A) DNA ligase
   (B) DNA polymerase
   (C) Helicase
   (D) Primase
8. Which law relates induced EMF to rate of change of magnetic flux?
   (A) Lenz's law
   (B) Faraday's law
   (C) Ampere's law

(D) Ohm's law

9. Which type of mutation involves change in single nucleotide?
   (A) Deletion
   (B) Insertion
   (C) Point mutation
   (D) Frameshift

10. Which organelle is involved in programmed cell death?
   (A) Lysosome
   (B) Mitochondria
   (C) Peroxisome
   (D) Ribosome

Section B: Fill in the Blanks (1 mark each)

11. For a reversible process, entropy change of the universe is ___.

12. Raoult's law applies to ___ solutions.

13. The magnetic field inside a solenoid is given by $B = \mu$■___.

14. The enzyme that converts prothrombin to thrombin is ___.

15. The Krebs cycle occurs in the ___ of mitochondria.

16. The uncertainty principle was given by ___.

17. In SN2 reactions, the rate depends on both ___ and nucleophile.

18. The process of splitting water during photosynthesis is called ___.

Section C: Short Answer Questions (2 marks each)

19. Describe electromagnetic induction and derive Faraday's law.

20. Derive the lens formula and explain its applications.

21. Explain hybridization with examples of sp, sp^2, and sp^3.

22. Explain Le Chatelier's principle with examples.

23. Explain the structure and working of a nuclear reactor.

24. Derive the expression for electric field due to a dipole.

25. Derive the equation for kinetic energy of gases from kinetic theory.